

CSCE 628 COMPUTATIONAL BIOLOGY · MODULE 01 · LECTURE NOTES

What Is Actually Measured

The biology you need, and the gap this course is about.

MODULE OUTCOMES

- State the molecular objects and the central dogma.
- Describe what the common measurement technologies output.
- Explain the gap between measurement and biological claim.
- Explain why ground truth is usually unavailable.
- State this course's position.

1 The objects

Object	Alphabet	What it is
DNA	A C G T	The stored sequence ; double-stranded and complementary, so a read may come from either strand — which matters for every mapping algorithm.
RNA	A C G U	A transcribed copy of a region ; it also folds and acts catalytically or regulatorily (Module 08).
Protein	20 amino acids	Translated from RNA in triplets ; folds into a three-dimensional shape that determines what it does (Module 09).
Gene	—	A region that is transcribed — and the definition is genuinely contested, which has practical consequences (see the callout).
Genome	—	The whole sequence of an organism ; roughly three billion bases in a human, of which the protein-coding part is a small percentage.

DNA → RNA → protein is the central dogma, and every arrow has exceptions — reverse transcription, RNA that is never translated, proteins that modify other proteins — **which is a good introduction to how biology's rules work**: they are strong regularities with important exceptions, rather than axioms. **The alphabet sizes are what make the algorithms behave as they do**: a four-letter alphabet means random matches are common, which is Module 03's whole problem.

WHICH IS ENOUGH BIOLOGY TO START, AND THE EXCEPTIONS MATTER LATER

Most of this course is string algorithms and statistics — so the biology you need is the vocabulary and the measurement model, both of which this module supplies, and the rest can be looked up when a specific question arises.

But the exceptions are where the errors live: genes overlap, are spliced into multiple distinct products, sit inside the introns of other genes, and are defined differently by different annotation projects — so 'the gene' is frequently ambiguous and two analyses can disagree without either being wrong.

And the reference genome is a construct — assembled from a small number of individuals, revised repeatedly, and not any particular person's actual sequence — which has direct consequences in Module 12 §2, where coordinates from one build silently mean something else in another.

So the rule for this course is to look up what you are assuming rather than to learn biology in advance — and the assumption to check first is always 'which version of which annotation', which is the single most common source of a result that cannot be reproduced.

2 What the instruments produce

SHORT-READ SEQUENCING

output: millions of 100-300 base fragments, each with per-base quality scores, from roughly random positions, with PCR duplicates and errors. NOT a genome. Assembly or mapping comes next (Module 05).

LONG-READ SEQUENCING

tens of kilobases per read, higher per-base error rate, far better across repeats. Different trade, same shape of output.

RNA SEQUENCING

reads drawn from transcripts, counted per gene. The count depends on expression AND on transcript length AND on the library preparation. Normalisation is not optional (Module 10).

MICROARRAYS AND MASS SPECTROMETRY

intensities and spectra -- analogue signals with their own calibration and saturation problems.

IN EVERY CASE THE OUTPUT IS COUNTS OR INTENSITIES, and the biological quantity is inferred from them.

In every case the output is counts or intensities, and the biological quantity is inferred — which is the gap this course keeps returning to (§3). Knowing what the instrument actually emits prevents most interpretation errors: an RNA-seq count is not a concentration, a read is not a position, and an intensity is

not an amount. Each of those four-word corrections has a literature behind it.

3 The gap

EVERY RESULT IN THIS FIELD IS A CHAIN OF INFERENCES, AND THE CHAIN IS USUALLY REPORTED AS A SINGLE STEP

'Gene X is upregulated in disease' means: reads were mapped to a particular annotation of a particular reference build, counted per feature, normalised by some method, modelled under some distributional assumption, and tested — five steps, each involving choices, applied to tissue from a particular set of people at a particular time.

And each step has defensible alternatives that change the answer — different aligners, annotations, normalisation methods, and statistical models produce overlapping but genuinely different gene lists from identical raw data.

Which is not a scandal — it is what analysis of noisy, high-dimensional data looks like — and it is a reason to report the chain rather than the conclusion alone, so that a reader can tell which choices the finding depends on.

So this course's habit is to name the chain, every time — which is CSCE 679 Module 13's five-clause caption and CSCE 676 Module 13's proxy problem, arriving in a field where the stakes are medical and the downstream reader is frequently not an analyst.

4 No ground truth

- A sorting algorithm's output is checkable in linear time; an inferred gene function is not checkable at all without an experiment that may take years and cost a great deal — which is a structural difference from every other algorithms course in this program.
- So the field validates by proxy: agreement with other computational methods, recovery of known positive controls, performance on held-out data, and eventual experimental follow-up — each of which is partial and none of which is the checkable answer.
- And agreement between methods is weak evidence when the methods share assumptions, which they usually do — two aligners built on the same scoring model agreeing tells you less than it appears to (CSCE 676 Module 11's multiplicity argument).
- Which makes negative controls disproportionately valuable — a permutation that should find nothing and does find nothing is the strongest cheap evidence available, because it tests the whole pipeline including the parts you did not think about.
- And it is why Project 2 requires one and fails without it — it costs one rerun with shuffled labels, and it catches a class of error that no amount of careful reasoning does (Module 13 §2).

SO THE POSITION THIS COURSE TAKES

The algorithms are the easy part, and they are worth learning properly — Modules 02 through 09 are genuinely beautiful dynamic programming and probabilistic modelling, and this course does not short them or treat them as background to the statistics.

And the inference is where the errors are — Modules 10 through 12 are about multiple testing, confounding, batch structure, and reproducibility, which is where published findings actually fail rather than where they are usually scrutinised.

Plus the honest acknowledgement: large fractions of published findings in several subfields have failed to replicate, and the causes are statistical and procedural rather than biological — which is simultaneously discouraging and encouraging, since procedural causes are fixable.

Which is why Module 13 exists — 'we found an association' is a claim about a statistical model applied to a particular cohort, and the distance between that and the biological claim is exactly what this course is trying to make visible.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
The NCBI Science Primer and handbooks (free)	§1 — the biology vocabulary, at the level this course needs and no more.
Scitable and the open genetics primers (free)	§1's exceptions — splicing, overlapping genes, and why 'the gene' is contested.
Illumina and PacBio technology overviews (free)	§2 — what the instruments do, from the manufacturers, read for the mechanism rather than the marketing.
Ioannidis — Why most published research findings are false (free)	§4 and the closing callout — the argument, which is statistical and applies directly to Modules 10 and 11.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Write out the central dogma** and find one exception to each arrow.
- 2 Look up one gene in two annotation sources** and compare their coordinates.
- 3 Find out which reference build** a dataset you can access used.
- 4 Download a FASTQ file** and look at the quality scores.

- 5 **Count the duplicate reads** in it.
- 6 **Describe the output of three technologies** in your own words.
- 7 **Take one published sentence of the form 'gene X is upregulated'** and enumerate the inference chain behind it.
- 8 **Find two papers analysing the same public dataset** and compare their gene lists.
- 9 **Design a negative control** for an analysis you might run.
- 10 **State what this course assumes about biology**, and what it does not.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the four objects and their alphabets.
- 2 State the central dogma, and why its exceptions matter.
- 3 Why is 'the gene' ambiguous, and what should you check first?
- 4 What is the reference genome, actually?
- 5 Describe what short-read sequencing outputs.
- 6 Why is RNA-seq normalisation not optional?
- 7 What is the one thing true of every technology's output?
- 8 Enumerate the five steps behind an upregulation claim.
- 9 Why is agreement between methods weak evidence?
- 10 Why are negative controls disproportionately valuable?